

Designing, Prototyping and Construction

Assistive Devices for those on the Spectrum

We are at the final leg of our assignment as we discuss one of the most crucial topics for our device – the Design, Prototype and Construction.

Design: It is imperative that the device is easy to use while being functional, We need to ensure that the icons on the App are big and clearly visible to the users and it is often that people who are mentally challenged often get frustrated they encounter a situation that they cannot comprehend. Another Design Perspective would be to ensure that the device would stay intact on the person at all times when it is worn and shouldn't be able to come off due to movement, in order to achieve this, we need to study the ergonomics of the movement of mentally challenged patients and design the device accordingly.

Key Features:

- Simplistic
- Light
- Portable
- Holistic
- Durable

Prototype 1: The prototype of this device can be made by utilizing a functional App that possess Bluetooth and Wi-Fi capability, along with a Bluetooth device. The setup must be in a manner such that when the pairing between the devices is interrupted the device must trigger and send an alert to some pre-determined devices that the pairing has been broken between the device and the Phone.

Prototype 2: The prototype can be constructed by imbedding an apparatus into a cavity of a pillow that on applying pressure (pressing it) must sound a noise that is clear and distinctive. This would represent the SOS feature to send out a distress call to alert the pedestrians nearby.

Construction: Having understood the crucial necessities of the Device it is Important that for the construction of the device we main and ensure that we achieve the key features of the device.

A lightweight material must be utilized for deploying the device in order to reduce the burden on the person who utilizes the device. Furthermore, the device must be constructed keeping in mind that all the features must be well integrated with the App to perform seamlessly and provide maximum User Experience. While, the design aspects may limit the creativity it is more crucial that ease of access is given priority over the prior.

Thus, I conclude the Assistive Device that I have explained in the series of assignments that I have submitted in the past couple of months. While, this device may not appeal to the mass public since it targets a specific market. The humanity aspect of this device cannot be overlooked since it will help improve and augment the way of life of those who are mentally challenged and can cover a wide range of patients who suffer from ADHD, PTSD, Down's Syndrome, Vertigo, etc.